INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, April – May, 2025

Degree: **B.Tech**, Semester: **Second**

Stream: **Biotech/CSE(AI)/CSE(AI & ML)/ CSE(AIML)/CSE(Cyber Security)/CSE(Data Science)/Robotics and AI/CST/ECE/ME/EE/EEE**

Paper Code: **BSCPH201**

Paper Name: **Physics**

Full Marks: **100** Duration: **3 Hours**

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 × 10)

1. What are scalar and vector fields? Give examples. **2**

2. Define divergence of a vector point function. What is its physical significance? **2**

3. State Gauss' divergence theorem. Explain the physical significance of this theorem. **2**

4. Write down the differential equation of an *LCR* citcuit. **2**

5. What are the conditions for overdamped, critically damped and underdamped motions? **2**

6. What are coherent sources? **2**

7. What is interference of light? **2**

8. Explain the colours when a thin film illuminated by white light is observed in reflected light. **2**

9. Write down Faraday's law of electromagnetic induction. **2**

10. Explain the significance of displacement current. **2**

11. Distinguish between the conduction current and displacement current. **2**

12. What is probability density? **2**

13. Can you normalize every wave function? **2**

14. Define thermodynamical probability. **2**

15. What are fermions and bosons? **2**

Part - B

Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Show that (i) curl grad $v = 0$ (ii) div curl $\vec{A} = 0$. **5**

17. State parallel axes and perpendicular axes theorem. **5**

18. Derive the equation of a simple harmonic oscillator form the energy consideration. **5**

19. State Rayleigh critertion of resolution and discuss its significance in studying spectral lines. **5**

20. The ratio between maximum and minimum intensities is 36:1 in an interference pattern. Find the ratio between the amplitudes and intensities of the two interfering waves. **5**

21. Write and explain differential and integral forms of Maxwell's equations. **5**

22. State Lenz's law. Explain it from the principle of conservation of energy. **5**

23. Draw the graphs of ϕ *and* $|\phi|^2$ for the first three states of a harmonic oscillator. **5**

24. Prove that at 0 K the average energy is $\frac{3}{5}\varepsilon_F$ **5**

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. A hollow sphere and a solid cylinder of same masses and same radius start from rest roll down a rough incline plane, without slippling. Find their accelerations after they have fallen through a certain distance. **10**

26. What are the similarities and dissimilarities of critically damped and overdamped motion? Show that the ratio of the successive amplitudes of damped oscillatory motion is constant. **10**

27. Distinguish between amplitude resonance and velocity resonance. Show that at velocity resonance, the maximum velocity is inversely proportional to damping and the velocity of the oscillator is in phase with the periodic external force in the steady state. **10**

28. Discuss the effect of introducing a thin plate in the path of one of the interfering beams in a biprism experiment. Deduce an expression for the displacement of the fringes. **10**

29. What are Newton's rings? Describe and explain the formation of Newton's rings in reflected monochromatic light. Prove that in reflected light, **10**

a) Diameters of the dark rings are proportional to the square root of natural numbers, and

b) Diameters of the bright rings are proportional to the square root of odd natural numbers.

30. Show that $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ and $\vec{\nabla} \times \vec{B} = \mu_0 \left[\vec{J} + \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right]$ **10**

31. Show that **10**

a) $[\hat{p}_x, \hat{x}_m] = -i\hbar n x^{n-1}$ and

b) $[\hat{p}_x, f(x)] = -i\hbar \frac{df}{dx}$

32. **10**

a) Compare MB, BE and FD statistics mentioning at least three characteristics.

b) Draw the Fermi distribution curve for (i) T= 0 K and (ii) T > 0 K. Explain their significance.

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